AJP PROJECT

Topic: Examination Portal

Team Members:

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ABSTARCT:

The Online Examination Portal is a web-based application designed to help students evaluate their knowledge in various subjects through structured online tests. The system provides role-based access for administrators and students, ensuring secure and efficient management of examinations.

The Admin module enables the administrator to manage student accounts, create and update tests, and add questions for different subjects. The Student module allows registered users to take exams, view individual and cumulative scores, and analyze their performance through graphical reports. The system also maintains statistics such as the number of attempts, average score, minimum score, and maximum score for each student.

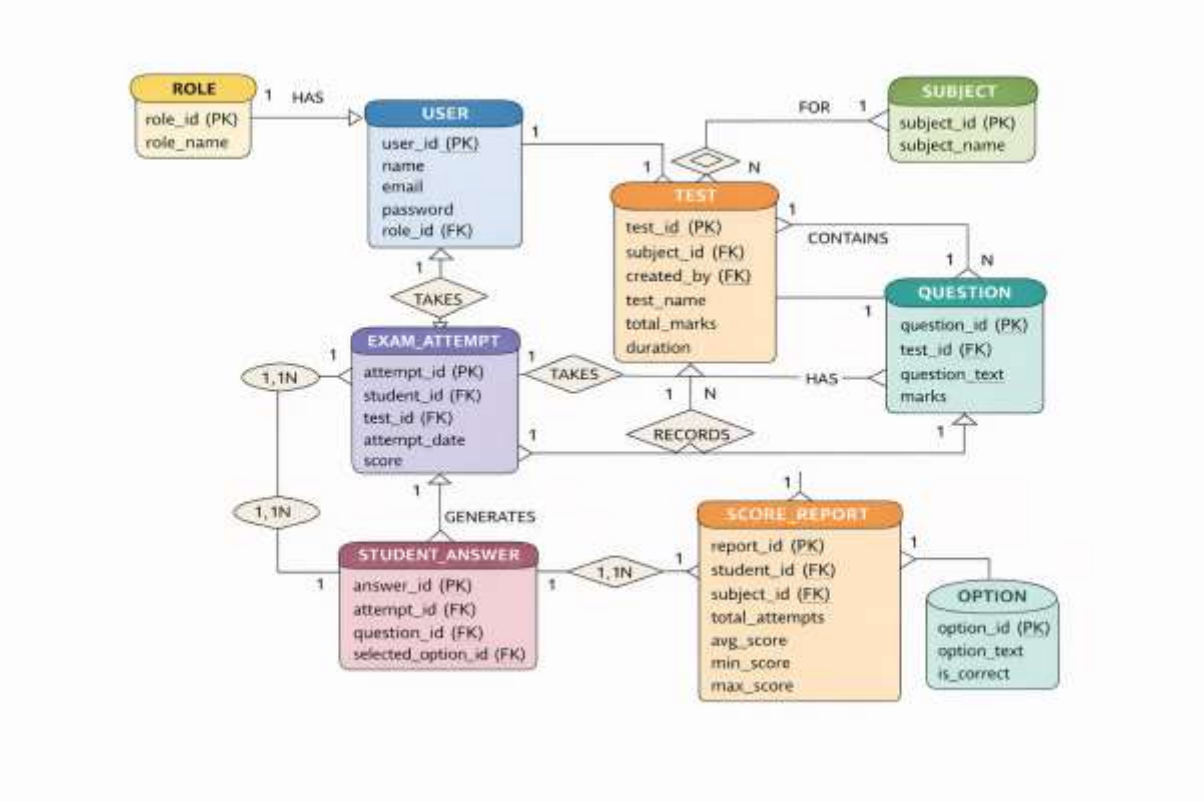
Additionally, the portal generates subject-wise and student-wise performance reports to help monitor academic progress. Students can email their score reports directly to their registered email IDs. The application is implemented using Spring Boot/Spring MVC with ReactJS (or JSP/Thymeleaf) and MySQL with Hibernate/JPA for efficient data persistence.

The system aims to provide a reliable, scalable, and user-friendly platform for conducting and managing online examinations.

ER DIAGRAM:

Main Entities

1. User
2. Role
3. Subject
4. Test
5. Question
6. Option
7. Exam_Attempt
8. Student_Answer
9. Score_Report



DATABASE DESIGN

Design Overview

The database is designed to:

- Support Admin and Student roles
- Manage subjects, tests, and questions
- Track student attempts and answers
- Generate performance statistics
- Support reporting and email features

Normalization: Up to Third Normal Form (3NF)

TABLE STRUCTURE

1. ROLE

Purpose: Stores user roles (Admin, Student)

Column Name	Data Type	Constraints
role_id	INT	PRIMARY KEY, AUTO_INCREMENT

Column Name	Data Type	Constraints
role_name	VARCHAR(20)	NOT NULL, UNIQUE

2. USER

Purpose: Stores all system users

Column Name	Data Type	Constraints
user_id	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(120)	NOT NULL, UNIQUE
password	VARCHAR(255)	NOT NULL
role_id	INT	NOT NULL, FK → ROLE(role_id)
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
status	VARCHAR(20)	DEFAULT 'ACTIVE'

3. SUBJECT

Purpose: Stores subjects

Column Name	Data Type	Constraints
subject_id	INT	PRIMARY KEY, AUTO_INCREMENT
subject_name	VARCHAR(100)	NOT NULL, UNIQUE
description	VARCHAR(255)	NULL

4. TEST

Purpose: Stores tests created by admin

Column Name	Data Type	Constraints
test_id	INT	PRIMARY KEY, AUTO_INCREMENT

Column Name	Data Type	Constraints
subject_id	INT	NOT NULL, FK → SUBJECT(subject_id)
created_by	INT	NOT NULL, FK → USER(user_id)
test_name	VARCHAR(150)	NOT NULL
total_marks	INT	NOT NULL
duration_minutes	INT	NOT NULL
is_active	BOOLEAN	DEFAULT TRUE
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

5. QUESTION

Purpose: Stores questions for each test

Column Name	Data Type	Constraints
question_id	INT	PRIMARY KEY, AUTO_INCREMENT
test_id	INT	NOT NULL, FK → TEST(test_id)
question_text	TEXT	NOT NULL
marks	INT	NOT NULL
difficulty_level	VARCHAR(20)	NULL

6. OPTION

Purpose: Stores options for MCQ questions

Column Name	Data Type	Constraints
option_id	INT	PRIMARY KEY, AUTO_INCREMENT
question_id	INT	NOT NULL, FK → QUESTION(question_id)
option_text	VARCHAR(255)	NOT NULL

Column Name	Data Type	Constraints
is_correct	BOOLEAN	DEFAULT FALSE

7. EXAM_ATTEMPT

Purpose: Stores each time a student takes a test

Column Name	Data Type	Constraints
attempt_id	INT	PRIMARY KEY, AUTO_INCREMENT
student_id	INT	NOT NULL, FK → USER(user_id)
test_id	INT	NOT NULL, FK → TEST(test_id)
attempt_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
score	INT	NOT NULL
total_questions	INT	NULL
time_taken_seconds	INT	NULL

8. STUDENT_ANSWER

Purpose: Stores answers selected by student

Column Name	Data Type	Constraints
answer_id	INT	PRIMARY KEY, AUTO_INCREMENT
attempt_id	INT	NOT NULL, FK → EXAM_ATTEMPT(attempt_id)
question_id	INT	NOT NULL, FK → QUESTION(question_id)
selected_option_id	INT	NOT NULL, FK → OPTION(option_id)
is_correct	BOOLEAN	NULL

9. SCORE_REPORT (Performance Summary)

Purpose: Stores aggregated statistics

Column Name	Data Type	Constraints
report_id	INT	PRIMARY KEY, AUTO_INCREMENT
student_id	INT	NOT NULL, FK → USER(user_id)
subject_id	INT	NOT NULL, FK → SUBJECT(subject_id)
total_attempts	INT	DEFAULT 0
avg_score	DECIMAL(5,2)	DEFAULT 0
min_score	INT	DEFAULT 0
max_score	INT	DEFAULT 0
last_updated	DATETIME	DEFAULT CURRENT_TIMESTAMP

MODULE MAPPING:

Admin Modules

1. Registration/Login Module
2. Students Updation Module
3. Tests Updation Module

Student Modules

4. Examination Module
5. Performance Module
6. Reports Module

TECH STACK :

- Frontend: ReactJS / Thymeleaf / JSP
- Backend: Spring Boot / Spring MVC
- ORM: Spring Data JPA / Hibernate
- Database: MySQL
- Build Tool: Maven
- Email: JavaMailSender